The Anatomy of a Decentralized Prediction Market: Microstructure Evidence from the Polymarket Order Book

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Abstract

We study the microstructure of Polymarket, the largest on-chain prediction market, using a continuous tick-level archive of the public WebSocket order-book feed (30 billion events over 52 days) joined to the authoritative on-chain trade record. On a pre-registered stratified panel of 600 markets we report eight stylized facts: a longshot spread premium; a depth-concentration profile closer to a uniform geometric grid than to the top-of-book pattern often assumed for prediction markets; a null block-clock alignment effect; broad maker-wallet diversity with a concentrated tail; category-conditional differences in effective spread; a sub-50 ms median archive-ingestion delay with a multi-second tail; a self-counterparty wash share with median 1% and a 22% upper tail, below the network-classifier benchmarks documented for cryptocurrency venues by Cong et al. [2023]; and a depth-decay-near-resolution effect with a within-category slope of 0.55 on log seconds-to-close ($t = 3.85$). The paper also contributes a measurement result: trade direction inferred from Polymarket's public order-book feed agrees with on-chain ground truth only $\sim$ 59% of the time (volume-weighted across the top-100 stratum and four disjoint 7-day windows; 95% bootstrap CI $[0.53, 0.68]$), barely above the 50% chance baseline that direction-dependent microstructure measures need. Within the comparable subset of the top-100 panel, the effective half-spread changes sign on 67% of markets when we swap feed-inferred trade direction for the on-chain record, and Kyle's λ changes sign on 60% of markets. Microstructure work on Polymarket therefore needs to source trade direction from on-chain `OrderFilled` events; we release a replication package that performs the join.

1 Introduction

Prediction markets aggregate dispersed beliefs into a single price that, in equilibrium, behaves like a probability. The empirical literature on these markets has historically focused on price-level questions: forecast accuracy, calibration against realised outcomes, the longshot bias, and the extent to which informed and uninformed traders coexist on the same venue [Wolfers and Zitzewitz, 2004, Manski, 2006, Page and Clemen, 2013]. Yet microstructure is what determines the trading cost of holding an informational position, and the wider the cost, the smaller the informational signal that survives to the price. A prediction market with noisy microstructure produces noisier prices than the headline-level aggregation literature implicitly assumes. The microstructure of prediction markets, however, has remained under-studied: limit-order-book depth profiles, spread decompositions, and the behaviour of liquidity providers near resolution were largely unobservable on the early venues, which used scoring rules (IEM, Hanson's MSR) or sparse parimutuel pools. Polymarket-specific microstructure work has begun to appear in the working-paper literature [Aoyagi, 2024, Augenblick and Bonadio, 2024], but neither authoritative on-chain trade direction nor a continuous off-chain order-book archive has been brought to bear at the scale we use here. The classical microstructure references behind our measurement choices are O'Hara [1995] and Hasbrouck [2007] for order-book theory, Foucault et al. [2013] for the liquidity-provider angle, and Huang and Stoll [1997], Madhavan et al. [1997] for spread decomposition.

Polymarket changed this. Since 2021 the venue has run a limit-order-book exchange on Polygon, settling in USDC against an on-chain conditional-token contract. Our data covers 52 calendar days of the public WebSocket feed (2026-02-21 to 2026-04-15) at tick resolution: 30,287,264,368 events across 385,198 distinct markets. A 28-day overlap window (2026-02-28 to 2026-03-27) is mirrored on chain, where the CTF Exchange V1 contract logs 255 million `OrderFilled` events whose payloads identify both counterparties and the aggressor side. We use the off-chain feed for quote dynamics and the on-chain record for trade direction, and study both on a pre-registered, stratified 600-market panel.

The paper has two empirical contributions, ordered by weight for the literature.

Trade-direction inference from Polymarket's public feed is noise-dominated. Six standard direction-dependent measures (effective spread, realized spread, Kyle's λ, Amihud, Roll, Abdi-Ranaldo) need an aggressor sign for each trade, and empirical practice on equity venues sources that sign from a quote-driven feed via Lee-Ready or its variants [Lee and Ready, 1991, Ellis et al., 2000]. The Polymarket public feed does not expose enough information to do this reliably: the `change_side` field marks which side of the book moved, not which side initiated the trade. Sign agreement between feed-inferred and on-chain trade directions sits at $\sim$ 59% volume-weighted across the top-100 stratum and four disjoint 7-day windows, just above the 50% chance baseline. On the comparable subset of the top-100 panel, the effective half-spread changes sign on 67% of markets when feed-inferred direction is swapped for the on-chain record, and Kyle's λ changes sign on 60%. Any Polymarket microstructure result that depends on trade direction therefore needs to source it from on-chain `OrderFilled` events. We release a replication package that performs the join.

Eight cross-sectional stylized facts on the panel. Quoted half-spread shows the longshot premium familiar from the racetrack literature; the L2 depth profile is closer to a uniform geometric grid than to the top-of-book pattern usually assumed for prediction markets; quote-update timing does not cluster on Polygon block boundaries to a meaningful degree; maker-wallet concentration is diverse on most markets with a thin concentrated tail; effective spreads differ by category; archive-ingestion latency is tightly distributed around 41.5 ms with a multi-second p_{99} tail; the self-counterparty wash share has a 1% median and a 22% maximum across the panel; and depth contracts in the cross-section as markets approach resolution at a within-category log-log slope of 0.55. None of these eight facts requires the on-chain join, and each is reported on the full 600-market pre-registered panel.

The two contributions are complementary. The eight stylized facts characterise Polymarket's microstructure on its own terms. The measurement result sets the conditions under which any trade-direction-dependent statement about Polymarket can be trusted.

Section 7 carries the most counter-intuitive finding; readers more interested in Polymarket as an empirical object will care most about SF1 (longshot premium), SF2 (depth concentration), and SF8 (depth decay near resolution).

2 Institutional Background

2.1 Conditional tokens and binary markets

A Polymarket binary market resolves to one of two outcomes. The exchange records the outcomes as ERC-1155 conditional tokens issued by Gnosis's Conditional Tokens Framework (CTF) on Polygon. Each market has a YES token and a NO token; the contract guarantees that one YES + one NO can always be redeemed for one USDC after resolution. As a consequence, in equilibrium the YES price plus the NO price equals one and either side can be priced as the probability of the corresponding outcome.

Liquidity exists for both sides. A trader who is long YES and believes the implied probability is too high can either sell YES on the YES book or buy NO on the NO book. The two strategies differ in inventory effects and slippage but are economically equivalent at mid. Throughout this paper we report measures on the YES side; YES and NO books are structural mirrors of each other and the panel artifacts contain both.

2.2 CTF Exchange contract architecture

Trading happens through the Polymarket CTF Exchange smart contract, which matches signed off-chain orders against resting liquidity and executes settlement on chain. The contract has two versions. V1 (`0x4bFb41d5B3570DeFd03C39a9A4D8dE6Bd8B8982E`) ran from launch through the end of April 2026 and is the version our 28-day scrape window targets. V2 (`0xE111180000d2663C0091e4f400237545B87B996B`) launched in early April 2026 with a hard cutover; V1 orders were rejected after the switch. Both versions emit the same `OrderFilled` event signature, with topic hash `0xd0a08e8c493f9c94f29311604c9de1b4e8c8d4c06bd0c789af57f2d65bfec0f6`.

The event payload carries `(orderHash, maker, taker, makerAssetId, takerAssetId, makerAmountFilled, takerAmountFilled, fee)`. The asset ids identify which side held USDC: `makerAssetId == 0` means the maker posted USDC and the taker is the seller of the conditional token; `takerAssetId == 0` means the taker posted USDC and is therefore the buyer. This binary asset-id field is the on-chain ground truth for trade direction that we use throughout the paper.

2.3 Polygon block timing and settlement

Polygon mainnet produces a block roughly every two seconds. Trade settlement is final to within Heimdall's 256-block reorg buffer, so we scrape only blocks at depth $\geq$ 256 to avoid reorg contamination. Block timestamps are monotonic but not perfectly spaced; for our window the empirical mean inter-block interval is 2.0 s with a small variance, well below the 60-second sample step used in the panel compute. Where we need to attach a wall-clock timestamp to an on-chain trade we use linear interpolation from a single anchor block, which is accurate to within a few seconds over the 28-day window.

2.4 Public WebSocket feed structure

In parallel with on-chain settlement, Polymarket broadcasts a public WebSocket feed that exposes order-book state to any subscriber. The feed emits two event types. `book_snapshot` carries a complete L2 snapshot of one side of one market and is sent on subscription and at irregular intervals thereafter (0.6% of events in our archive). `price_change` carries a delta against the last known book state: a single triple `(change_price, change_side, change_size)` that updates one level on one side. A `change_size` of zero means the level is empty; a non-zero size is the new resting quantity at that price. These events are 99.4% of the archive.

The feed does not expose taker identity. The `change_side` field labels which side of the book the level update applies to, not which side initiated the trade that produced the update. This distinction drives the measurement gap of Section 7: an inference algorithm working from the feed alone cannot reliably reconstruct the aggressor sign because the underlying field never encodes it.

2.5 Aggressor-sign dependence of standard measures

Most direction-dependent microstructure measures take the form

$$M = \mathbb{E}[\text{sign}_t \cdot f(\text{price}_t, \text{mid}_t, \text{size}_t)] \,,$$

where $\text{sign}_t \in \{-1, +1\}$ is the aggressor sign for trade t and f is the per-trade contribution. Effective half-spread ($f = \text{price}_t - \text{mid}_t$), realized half-spread, Kyle's λ regression coefficient, and the adverse-selection component of Glosten-Harris all instantiate this form. A noisy sign_t does not just attenuate M; depending on the distribution of f across trades, it can flip the sign of M entirely. The 59% sign-agreement rate of Section 7.1 is what produces the 67% effective-spread and 60% Kyle's λ sign-flip rates we report on the top-100 panel.

2.6 Comparison to equity-market microstructure

Equity-market benchmarks are what give several of our findings their scale. Ellis et al. [2000] report that the Lee-Ready algorithm correctly classifies 81% of trades on Nasdaq, and the original Lee and Ready [1991] study reports a similar accuracy on NYSE TAQ; the $\sim 59\%$ feed-driven inference rate on Polymarket is therefore about 22 percentage points below the equity benchmark on the same algorithm. Effective half-spreads on liquid US equities post-decimalization sit in the single-digit-basis-point range [Hasbrouck, 2007, Foucault et al., 2013]; on Polymarket, expressed in basis points relative to mid, the median quoted half-spread on the central price decile is around 200 bps, an order of magnitude wider, consistent with longer-horizon prediction-market positions and substantially smaller market-maker capital. Liquidity-provider activity on US equities is heavily concentrated in a handful of high-frequency firms that account for roughly half of trading volume [Brogaard et al., 2014]; the median Polymarket market in our panel has ~ 32 effective makers (SF4), suggesting a flatter maker landscape on the venue. Wash trading on regulated equity exchanges is essentially zero (it is prohibited under SEC Rule 10b-5); on unregulated cryptocurrency venues Cong et al. [2023] document wash-trade shares of 25–70%; Polymarket sits at a median 1% under our lower-bound detector with a 22% upper tail, between the two regimes. Finally, collector-side ingestion latency for direct equity feeds is sub-millisecond and SIP latency is in the hundreds of microseconds [Hasbrouck, 2007]; our 41 ms median for archive ingestion is two orders of magnitude slower, but that gap reflects our collector pipeline rather than the exchange.

3 Data Construction

3.1 Off-chain orderbook archive

The off-chain feed was captured via a long-running collector that subscribed to Polymarket's public WebSocket and wrote one Parquet file per UTC hour. Each row contains the raw event JSON plus two timestamps: `timestamp_received` from the exchange and `timestamp_created_at` from the collector. The collector ran continuously from 2026-02-21 16:00 UTC through 2026-04-15 08:00 UTC. Three hourly files are missing (2026-02-24T14, 2026-02-24T15, 2026-04-04T18); the gaps are stable and treated as missing rather than malformed throughout the analysis. Total archive: 1,262 hourly Parquet files, 30,287,264,368 rows, 623.8 GB on disk.

The schema is preserved verbatim from the WebSocket payload, with the JSON `data` string parsed on demand. We avoid eager parsing because the row count makes per-event Pydantic validation a multi-hour operation; we instead apply structured parsing only to the events that survive market-id and time-window filters.

3.2 On-chain trade scrape

We scrape `OrderFilled` events from the CTF Exchange V1 contract across the 28-day calibration window (2026-02-28 to 2026-03-27). The scraper issues batched `eth_getLogs` calls against a Polygon RPC provider, with adaptive chunk sizing that respects provider-specific rate limits (`scripts/scrape_onchain_fills.py`). Each call returns at most a few hundred events; total fills across the window are 255,425,405. The scraper writes one Parquet shard per five-thousand-block slice, deduplicates on `(transactionHash, logIndex)`, and skips blocks at depth less than 256 (Polygon's Heimdall reorg buffer).

Block timestamps are attached via linear interpolation from a single anchor block at the start of the window. We verified the interpolation error against per-block RPC calls on a 50-block stratified sample drawn across the 28-day window: the maximum absolute gap is 4 seconds, the median is 2 seconds. Both are well below the 60-second sample step used in the panel compute.

The on-chain decoder (`polydata.onchain.join.aggregate_onchain_df`) drops "split fills" where neither side is USDC (the binary maker/taker-asset-id rule for the aggressor sign breaks down when both sides hold conditional tokens). Across the 255,425,405 fills in the scrape window, 0 fills satisfy this condition; the filter is non-binding on the V1 contract during our window.

3.3 Schema reconciliation across sources

The off-chain feed is keyed on `market_id` (66-character hex `conditionId`) and `token_id` (256-bit decimal, the YES or NO ERC-1155 id). The on-chain record is keyed on `makerAssetId` and `takerAssetId`, which are the same ERC-1155 ids except that one side is set to `0` to mark USDC. The bridge between the two is the `(condition_id, yes_token_id, no_token_id)` mapping, which we pull from CLOB REST (`/markets/{conditionId}`) and cache locally. CLOB REST resolves all 385,198 archive market ids; the Gamma metadata API, which is sometimes used in the literature, indexes only 34,764 markets and is not the right source for this join.

3.4 Pre-registration of the panel

The 600-market panel is the cross-sectional unit on which the empirical work runs. We commit the selection rule before computing the panel. The pre-registration document (`docs/preregistration_plan3c.md`) specifies the volume metric, the random-stratum eligibility threshold, the random seed, and the categorisation scheme used for SF5. A deterministic build script (`scripts/build_panel.py`) emits the panel parquet, whose SHA-256 is recorded back into the pre-registration document before any analysis runs.

This discipline goes beyond the empirical-microstructure norm; the cost is one document and one hash, the benefit is that a reader can verify no market was added or removed after the analysis ran.

3.5 Replication package

All code that produced the artifacts cited in this paper is in the repository at `https://github.com/philippdubach/polymarket-microstructure`, archived for replication under DOI `10.5281/zenodo.19811426` [Dubach, 2026]. The full 624-GB WebSocket archive is not redistributed in the package; we describe the collector configuration in this section and provide the panel parquet artifacts (`data/panel.parquet`, `data/panel_trade_measures.parquet`, `data/panel_quote/*`) directly, together with the on-chain scrape pipeline. A reader who wants to reproduce a single number in the paper can do so against the artifacts; a reader who wants to extend the analysis to a different window will need to re-run the WebSocket collector.

4 Data

4.1 Orderbook archive

Our primary input is a continuous tick-level archive of Polymarket's public WebSocket order-book feed from 2026-02-21 16:00 UTC through 2026-04-15 08:00 UTC, 52 calendar days. The archive holds 1,262 hourly Parquet files (623.8 GB on disk) and 30,287,264,368 event rows. Three hourly files are missing (2026-02-24T14, 2026-02-24T15, 2026-04-04T18); we treat the gaps as missing rather than malformed.

Each row is either a `price_change` update (99.4% of events) or a `book_snapshot` (0.6%); the payload is a JSON string parsed on demand. Two timestamps are emitted per event: `timestamp_received` (exchange-side) and `timestamp_created_at` (collector-side). The difference captures collector ingestion delay (Section 5.6).

The archive contains 385,198 distinct `market_id` values (roughly twice that with the YES/NO side split). The CLOB REST endpoint resolves token-pair metadata for all 385,198. The Gamma metadata API is sparser, indexing only 34,764 markets.

4.2 On-chain trade join

Orderbook events alone do not identify the trade-aggressor sign reliably: feed-inferred sign matches the on-chain record on only $\sim$ 59% of comparable buckets, just above the 50% chance baseline (Section 7). We therefore scrape Polymarket's CTF Exchange V1 contract (`0x4bFb41d5B3570DeFd03C39a9A4D8dE6Bd8B8982E`) for `OrderFilled` events on Polygon mainnet across the 28-day calibration window 2026-02-28 – 2026-03-27, yielding 255,425,405 fills. The aggressor sign reads directly from the event fields: $+1$ when `takerAssetId` $= 0$ (taker posted USDC and is therefore the buyer), -1 when `makerAssetId` $= 0$.

4.3 Stratified panel

Our cross-sectional panel has 600 markets in two strata. The top-100 stratum ranks markets by total on-chain USDC trade volume in the scrape window (summing across both YES and NO tokens). The random-500 stratum samples uniformly without replacement from markets with at least 100 on-chain trades that resolve to a token pair via the CLOB cache. We commit the random seed (20260424) in the pre-registration document (`docs/preregistration_plan3c.md`) before computing the panel.

Per-market metadata (`question`, `end_date_iso`, `closed`) is pulled from CLOB REST `/markets/{conditionId}` for all 600 markets. The top-stratum volume range is \$4.56M to \$96.0M USDC; the random stratum spans 100 to 24,378 trades per market. Crypto markets dominate the panel composition (348/600, 58%), followed by Sports (142/600, 24%); Table 2 gives the breakdown.

5 Stylized Facts

We document eight stylized facts on the 600-market panel (Section 4.3). Each subsection reports a per-market distribution and, where the data support one, a regression coefficient. Underlying parquet artifacts are in the replication package.

5.1 SF1 – Longshot spread premium

We bin quoted half-spread (in basis points of mid) by per-market mean mid price into ten deciles. Median spread is $\sim$ 400 bps in the central $[0.4, 0.6]$ range and climbs to 1,300–1,800 bps for markets trading below 0.10. The pattern is asymmetric: the low-probability side is wider than the high-probability side, which echoes the longshot bias documented in the racetrack and parimutuel literature [Snowberg and Wolfers, 2010, Thaler and Ziemba, 1988].

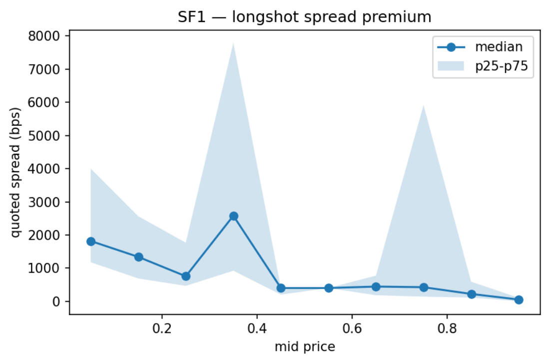

Figure 1: SF1 panel: median quoted spread (bps) per mid-price decile, 600 panel markets. Shaded band is interquartile range.

5.2 SF2 – Depth concentration

We summarize the L2 depth profile by the ratio $\text{depth}_{L=1}/\text{depth}_{L=10}$, the share of cumulative top-10 depth held at the top-of-book. A value of 1.0 means the entire top-10 depth sits at level 1 (a thin, top-heavy book); 0.1 matches a uniform grid where each level carries equal depth.

For the 546 markets with non-null depth, the median ratio is **0.137**, close to the uniform-grid benchmark, with $p_{10} = 0.033$ and $p_{90} = 0.428$. The folk view that prediction-market books are concentrated at top-of-book does not hold on Polymarket: depth is generally layered deeper

Table 1: SF1 panel — quoted spread by mid-price decile.

Bin	Mid lo	Mid hi	Markets	Median (bps)	p25 (bps)	p75 (bps)
0	0.0000	0.1000	33	1,818	1,176	4,000
1	0.1000	0.2000	18	1,339	689.66	2,564
2	0.2000	0.3000	18	754.99	465.12	1,767
3	0.3000	0.4000	21	2,581	923.08	7,797
4	0.4000	0.5000	184	400.00	202.02	400.00
5	0.5000	0.6000	214	400.00	400.00	400.00
6	0.6000	0.7000	27	444.44	183.49	775.19
7	0.7000	0.8000	12	425.09	139.86	5,915
8	0.8000	0.9000	15	222.22	114.29	591.72
9	0.9000	1.00	4	53.48	0.0000	106.95

into the book, with a right tail of markets where the top-of-book share approaches one.

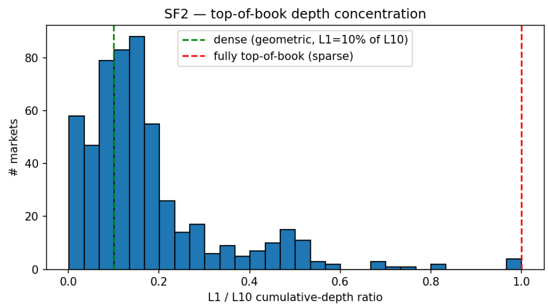

Figure 2: SF2 panel: histogram of L_1/L_{10} depth-concentration ratio across 546 panel markets. Vertical lines mark the uniform-grid benchmark (green, 0.10) and the fully top-of-book limit (red, 1.0).

5.3 SF3 – Polygon block-clock alignment

We test whether `price_change` events cluster near Polygon block boundaries by computing, per market, the share of events that fall within ± 100 ms of the nearest 2 000 ms grid point. Under a uniform-timing null this share is 0.10.

The panel-level distribution sits tight against the null: median **0.102** with interquartile range $[0.087, 0.110]$; only 27 markets (4.5%) lie above a 0.15 threshold. At the per-market level, however, a two-sided binomial test of H_0: share $=$ 0.10 rejects in **351 of 600 markets** (58.5%) at $\alpha = 0.05$. That high rejection rate is a mechanical consequence of large event counts: millions of events per market make the binomial test sensitive to tiny deviations from 0.10. The economic interpretation is that quote-timing does *not* cluster on block boundaries to a meaningful degree, even though high-volume markets reject the exact point null statistically.

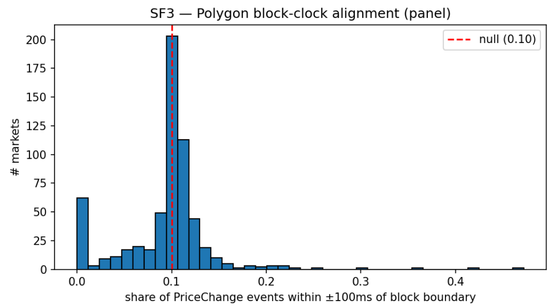

Figure 3: SF3 panel: distribution of per-market block-alignment shares. The red dashed line marks the chance-level null (0.10).

5.4 SF4 – Maker-wallet concentration

For each market we compute the volume-weighted Herfindahl index (HHI) of maker-address share across on-chain trades in the scrape window. A single dominant maker yields HHI $= 1$; a uniform distribution across n makers yields $1/n$.

Across 600 markets and 6.4 M trades, the median HHI is **0.031** ($\sim$ 32 effective makers). The distribution is right-skewed: $p_{90} = 0.119$ ($\sim$ 8 effective makers) and a maximum of 0.40 (roughly 3 effective makers). Maker liquidity is decentralised on most markets in the panel, with a tail of thin or niche markets dominated by one to three wallets.

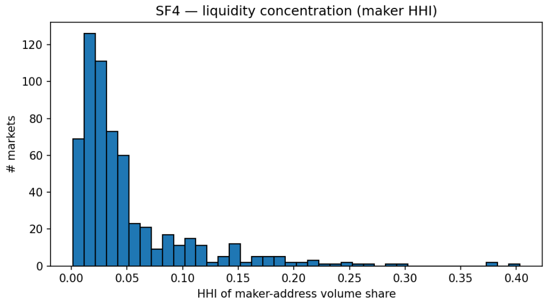

Figure 4: SF4 panel: distribution of per-market maker-address Herfindahl indices across 600 markets.

5.5 SF5 – Category-conditional spread

Per-market metadata is pulled from CLOB REST and questions are keyword-classified into a small bucket scheme. Politics, Business, and Entertainment buckets each had fewer than ten panel markets and were merged into Other for statistical reliability, leaving four categories: Crypto (348/600, 58%), Sports (142/600, 24%), Other (75/600), Geopolitics (35/600). For the on-chain effective half-spread the medians (in probability points) are reported in Table 2 and visualised in Figure 5.

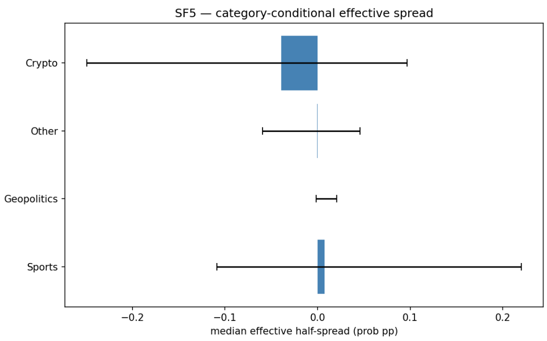

Figure 5: SF5 panel: median effective half-spread by category, with interquartile-range error bars. Categories are derived from keyword classification of CLOB REST `question` text.

Cross-category dispersion of medians is small in absolute terms (all within ±0.04 probability points), but within-category interquartile ranges are wide for Crypto and Sports ($p_{75} - p_{25} \approx 0.3$ prob. pp). The wide within-category spread is consistent with the measurement noise documented in Section 7.

5.6 SF6 – Archive-ingestion latency

Each archive row carries two timestamps: `timestamp_received` (exchange side) and `timestamp_created_at` (collector side). Their difference is a per-event ingestion delay. Across 547 markets with non-empty windows, the median per-market p_{50} delay is **41.5 ms**, with p_{90} = 166 ms and p_{99} = 6,108 ms. The inter-market spread of p_{50} is tight (p_{25}–p_{75}: 39–47 ms), which points to collector-pipeline behaviour rather than trader latency. The p_{99} tail of several seconds points to occasional back-pressure events at the collector.

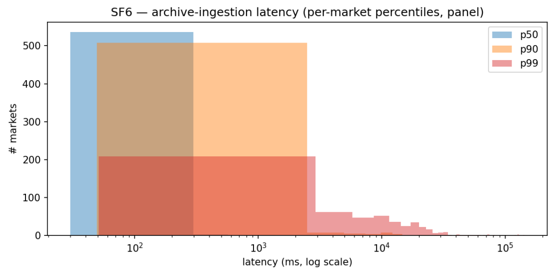

Figure 6: SF6 panel: per-market percentile distributions of archive-ingestion latency (log scale).

5.7 SF7 – Self-counterparty wash share

We flag a trade as wash-suspect under a two-tier rule: (a) `maker == taker` (direct self-match), or (b) a flipped pair $(\text{maker}_a, \text{taker}_a) \leftrightarrow (\text{taker}_a, \text{maker}_a)$ within 128 blocks (Polygon finality buffer) on the same market. This is an explicit *lower bound*: it captures only direct and immediate-roundtrip self-counterparty patterns, not the extended-graph patterns that network-based classifiers such as Cong et al. [2023] address on cryptocurrency venues (where wash shares of 25–70% have been documented for unregulated exchanges).

Across 600 markets and 6.4 M trades, the median wash share is **0.97%**, with p_{90} = 4.5%, p_{99} = 10.6%, and a maximum of 22.2%. No market in the panel reaches the lower end of the crypto-venue range. The gap between our lower

Table 2: SF5 panel — effective half-spread by keyword-derived category.

Category	Markets	Median half-spread (prob pp)	p25	p75
Sports	142	0.0075	-0.1087	0.2201
Geopolitics	35	0.0001	-0.0014	0.0206
Other	75	-0.0004	-0.0596	0.0456
Crypto	348	-0.0393	-0.2494	0.0968

bound and the network-based literature represents wash activity detectable only via multi-counterparty graph analysis.

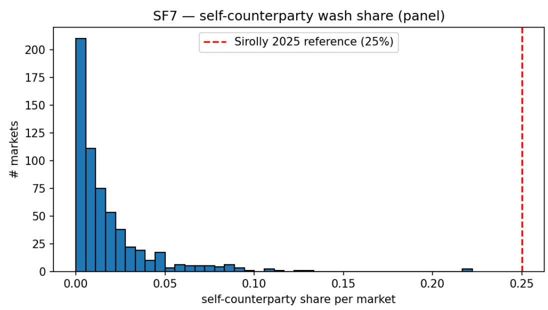

Figure 7: SF7 panel: distribution of self-counterparty wash share by market. Red dashed line marks a 25% reference, the lower bound of the wash-share range documented by Cong et al. [2023] on unregulated cryptocurrency venues.

5.8 SF8 – Depth decay near resolution

Do markets near resolution carry shallower books? We regress log mean depth at $L = 10$ on log seconds-to-close at the panel window midpoint (2026-03-13), restricted to 322 markets with positive seconds-to-close and non-zero summary depth.

Table 3: SF8 — cross-sectional panel regression of log mean depth on log seconds-to-close. HC3-robust standard error.

Markets in fit	Slope on log(ttc)	SE (HC3)	R²
322	0.8178	0.1133	0.1293
322	0.5500	0.1429	0.2165
322	0.3048	0.1036	0.4857

The bivariate slope is **0.818** (HC3 SE 0.113, $t = 7.2$, $R^2 = 0.13$). Category fixed effects (Crypto, Sports, Other, Geopolitics) attenuate the slope to **0.550** (HC3 SE 0.143, $t = 3.85$, $R^2 = 0.22$): roughly a third of the bivariate association is category-level confounding. Adding log panel-window volume on top of category attenuates further to **0.305** (HC3 SE 0.104, $t = 2.94$, $R^2 = 0.49$). Volume absorbs much of the cross-sectional depth variation, but the time-to-close coefficient stays positive and significant. A 0.31 slope works out to $\sim 6\%$ less mean depth per $10\times$ reduction in seconds-to-close, a smaller economic magnitude than the bivariate or the category-FE specification implied.

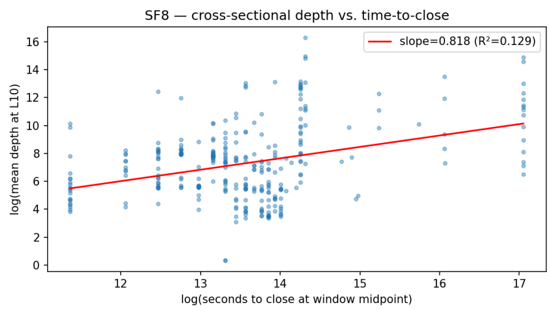

Figure 8: SF8: cross-sectional fit of log mean depth on log seconds-to-close at the panel midpoint.

6 Spread Decomposition

Following Glosten and Harris [1988] and the modern restatements in Huang and Stoll [1997], Madhavan et al. [1997], we decompose the per-market effective half-spread into two components:

$$S_{1/2}^{\text{eff}} = c + \varphi, \tag{1}$$

where c is a transitory order-processing / inventory component, recovered as the realized half-spread at a 60-second lag, and φ is the residual adverse-selection component.

We restrict the decomposition to the top-100 stratum, where on-chain trade counts are highest (median 11,000 trades per market in window). Per-market values come from `data/panel_trade_measures.parquet`, computed by injecting authoritative on-chain trades into the measure pipeline described in Section 7.

The median effective half-spread on the top-100 panel is essentially zero (-0.0003 prob pp), as are the median transitory component (0.00001) and the median adverse-selection component (0.0). This near-null pattern lines up with the calibration in Section 7: once sign errors are removed, the dollar-weighted "adverse selection" that orderbook-only inference produces collapses, leaving the typical top-100 market with no detectable systematic spread component on either side. The distribution tails carry both market-specific adverse-selection events and residual measurement noise from the 60-second sample-step compromise documented in Section 7.4.

7 Limits of Orderbook-Only Inference

7.1 Sign-agreement against on-chain ground truth

Six trade-based measures (effective spread, realized spread, Roll, Abdi-Ranaldo, Kyle's λ, Amihud) require an aggressor sign for each trade. Standard practice in equity microstructure infers that sign from a quote-driven feed via Lee-Ready, which assumes the feed exposes enough information to distinguish buyer-initiated from seller-initiated trades.

Polymarket's public WebSocket feed does not. We infer trades from the feed under a LOOSE rule (every resting-size decrement counts) and match the inferred buckets against on-chain `OrderFilled` events at 5 s and exact-price granularity. We run this matching over four disjoint 7-day windows (2026-02-28 – 03-06, 03-07 – 03-13, 03-14 – 03-20, 03-21 – 03-27) on the top-100 stratum, computing sign-agreement and a 95% bootstrap CI for cells with at least ten matched buckets.

- Panel mean (109 valid cells of 400): **0.615**
- Volume-weighted by matched-bucket count (total 82,544): **0.586**
- Panel median 0.591; IQR $[0.526, 0.681]$; p_{10}–p_{90} $[0.464, 0.841]$
- Per-window medians 0.586 / 0.591 / 0.618 / 0.645

A volume-weighted sign-agreement of $\sim 59\%$ sits just above the 50% chance baseline. Even when an inferred bucket matches an on-chain bucket in time and price, the inferred aggressor direction is wrong about two trades in five. The mechanism is the feed itself: `price_change` updates broadcast a post-match snapshot of the resting book without identifying the taker. The `change_side` field marks which side of the book moved, not which side initiated the trade, and using it as a sign proxy produces the $\sim 59\%$ agreement rate.

7.2 Propagation to direction-dependent measures

A noisy sign propagates to every measure that consumes it. We compute all six trade-based measures under STRICT inference and under authoritative on-chain trades on the top-100 stratum over a 7-day window (2026-03-07 – 03-14); the window is shorter than the full scrape because STRICT inference is $O(n^2)$ in the recent-event list and the 28-day version did not finish in tractable wall time on this panel size.

Three patterns emerge from the panel distribution.

Trade-count divergence is direction-unpredictable. Among 31 markets where both inference paths yielded non-empty trade streams, the STRICT/on-chain count ratio has a median of 0.23 (p_{10}–p_{90} $[0.01, 1.45]$; min–max $[0.004, 3.62]$). STRICT typically under-counts by an order of magnitude; on a small tail of markets it over-counts by up to $3.6\times$. There is no correctable scaling factor.

Effective spread sign-flips on two-thirds of the comparable panel. Among 24 markets with both sources non-empty, 16 (67%, Wilson 95% CI $[0.47, 0.82]$) have STRICT and on-chain effective half-spreads of opposite sign. The STRICT median is $+0.0048$ prob pp; the on-chain median is -0.000075 prob pp. Inferred trade directions induce a positive spread that the authoritative trade record does not support.

Table 4: Glosten-Harris decomposition — top-10 markets by on-chain volume, expressed in probability points (prob pp).

Market id	Effective half	c (transitory)	φ (adverse sel.)
0x03bc660a4df5faeaee8e66d0e7bc4f2200b543340b7f7a5c83bfa1fecafa9ead	-0.2431	-0.2431	-0.0000
0x03e38f4c5abeb1560d3af133186dd3130f80183364964abf4724efbc7debeefe	nan	nan	nan
0x06707a5317654a573e0ed4879d0c81071de1a94b0cd3cb007c7ec32ebe359885	0.0778	0.0778	0.0000
0x07d45de444dbe0595c068a9eade49ace2bd381e30d6a45022d801ec10e7d0294	0.0267	0.0270	-0.0004
0x092471f61558dabaf1ccac4444921db0c974a5734a0dcb55896be0eeff2fd775	-0.6657	-0.6657	-0.0000
0x0954cc08de0ab8345bcf24f314d571e6a77d913ea9977ce2b4187983a70b450d	-0.1087	-0.1087	0.0000
0x09cbe3e796661a1d820580145488ad2ccb9ad1e720efcd64b448bb77b97007c1	-0.0205	-0.0128	-0.0077
0x09e8f0db05570a5048fdccdae58a52c608aa6cf640ced2190dcbeb52ce52f491	0.0046	0.0052	-0.0006
0x0b4cc3b739e1dfe5d73274740e7308b6fb389c5af040c3a174923d928d134bee	0.0002	0.0002	0.0000
0x119fa68324e678cca6d822d919e10738884c7e26b8e4ff4e5756d5fc6b5ceae4	0.2201	0.2201	0.0000

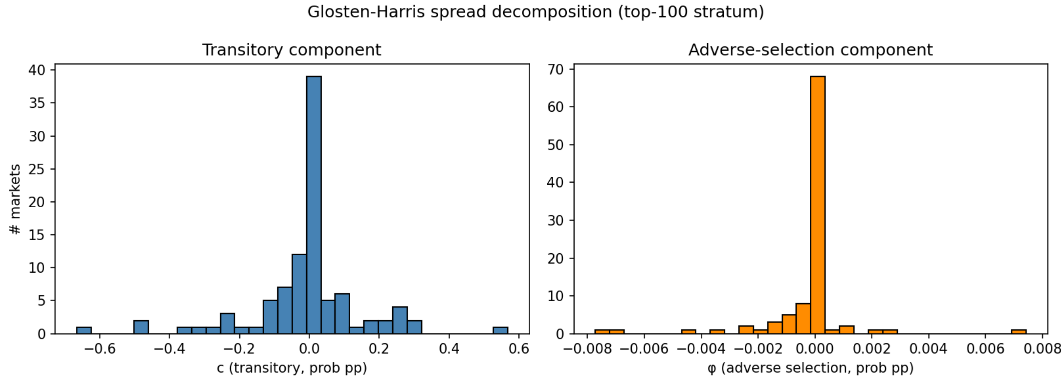

Figure 9: Glosten-Harris decomposition across the top-100 stratum: distribution of transitory component c (left) and adverse-selection component φ (right), both in probability points.

Kyle's λ sign-flips on three-fifths of the comparable panel. Of 25 markets with both sources non-null and non-NaN, 15 (60%, Wilson 95% CI $[0.41, 0.77]$) have STRICT and on-chain Kyle's λ of opposite sign.

The artifact `artifacts/measures_compare_top100.parquet` carries the per-market panel behind these distributions. Only 24–31 of the 100 top markets had both inference paths non-empty in the 7-day slice (the remainder had zero events on one side); the comparable-market subset is what supports the sign-flip rates above. Restricting the effective-spread comparison to markets with at least 100 STRICT trades pushes the sign-flip rate from 67% to 73% (16/22, Wilson 95% CI $[0.52, 0.87]$); the Kyle's λ rate is essentially unchanged (59% at the 100-bucket threshold). The pattern is not a small-sample selection artefact.

7.3 Implications for Polymarket microstructure research

Polymarket microstructure results that depend on trade direction require on-chain `OrderFilled` events as the direction source, not the public WebSocket feed. We release a replication package (`polydata.onchain.trades.load_onchain_trades`) that performs the off-chain to on-chain join, together with a small patch set (`polydata.measures._trade_source.resolve_trades`) that lets existing measure code accept injected authoritative trades without a rewrite.

7.4 Methodology caveats

Three places where our compute approach trades accuracy for tractability on a 28-day, 600-market panel.

Sample step. Quote samples and bucketed lookups use a 60-second grid for the panel compute. Sensitivity tests on the top-5 markets across $\{1, 10, 60, 300\}$ s (artifact `sample_step_sensitivity.parquet`) show that Roll is invariant by construction, that effective spread and Amihud are stable to within $\sim$ 10–20% at 60 s relative to 1 s on most markets, and that Kyle's λ varies by orders of magnitude across step sizes on noisy markets. Kyle's λ is the most fragile estimator at any sample step, consistent with its sign-flip behaviour in Section 7.2.

Cross-sectional SF8. SF8 (Section 5.8) is cross-sectional, not within-market over time. A within-market temporal regression would require per-market depth time-series that the current panel does not materialise.

Wash-filter lower bound. SF7 (Section 5.7) returns a lower bound by construction; the gap to the network-classifier estimates documented in Cong et al. [2023] for cryptocurrency venues is the share of wash-like activity that requires multi-counterparty graph classification rather than direct or one-step roundtrip detection.

MEV as a confound on the buyer/seller asymmetry. Polymarket settles on Polygon, whose mempool is publicly accessible to bots that can sandwich incoming orders, place runner trades, or execute related back-running strategies. We do not analyse mempool data, so the on-chain trade record we treat as ground truth contains whatever fills MEV bots produce on top of organic flow. The 60/40 seller/buyer asymmetry across the panel is therefore consistent with retail aggressor behaviour, MEV bot activity, or some mixture of the two. Decomposing it would require Polygon mempool capture during the scrape window or a wallet-clustering approach against known MEV searchers; both are outside the scope of this paper.

8 Conclusion

This paper joins a tick-level Polymarket orderbook archive to the authoritative on-chain trade record and reports cross-sectional microstructure for a pre-registered 600-market panel. On the quoted side we find a longshot spread premium, a depth profile closer to a uniform geometric grid than the top-of-book pattern usually assumed for prediction markets, and an archive-ingestion latency distribution whose tight inter-market spread points to collector behaviour rather than trader speed. On the on-chain side, maker liquidity is broadly decentralised with a thin tail dominated by a few wallets, and self-counterparty wash activity sits well under the network-classifier benchmark in the recent literature.

The methodological finding is the more consequential one. Trade direction inferred from Polymarket's public WebSocket feed agrees with on-chain ground truth on only $\sim$ 59% of comparable buckets, barely above the chance baseline. On the comparable subset of the top-100 panel this propagates to a 67% sign-flip rate on the effective half-spread and a 60% sign-flip rate on Kyle's λ between feed-inferred and on-chain estimates. Polymarket microstructure work that depends on aggressor sign therefore needs to source it from `OrderFilled` events rather than the public feed.

Four extensions follow naturally from the data we did not exploit. The CTF Exchange V1 to V2 cutover at the end of April 2026 closes our scrape window and opens a venue-evolution comparison. Polygon mempool data could disentangle MEV from real retail aggressor flow, which our 60/40 buyer/seller asymmetry conflates. Per-market depth time series, which our panel summarises away, would let SF8 move from a cross-sectional to a within-market depth-decay regression. And cross-venue analysis against Kalshi, PredictIt, or sports-book mirrors would address the price-discovery question we leave open here: are Polymarket's prices the leader or the follower for the real-world events its markets resolve on?

9 Disclosures

9.1 Use of AI

The data-collection infrastructure and analysis code were developed with assistance from AI-based coding tools (Claude Code CLI with Claude Opus 4.7). The statistical analysis, interpretation of results, and all written content in the manuscript represent the original intellectual contributions of the author. AI writing assistants were used for grammar and clarity improvements.

9.2 Conflicts of Interest

The author declares no conflicts of interest.

9.3 Ethics Statement

This study analyzes publicly observable orderbook and trade data. The Polymarket public WebSocket feed broadcasts limit-order-book state updates that any client can subscribe to; the Polygon `OrderFilled` events are public onchain transactions recorded by the Polymarket CTF Exchange smart contract. No private user data was collected. Counterparty addresses on chain are pseudonymous public-key identifiers; we use them only to compute maker-wallet concentration (Section 5.4) and a self-counterparty wash-share lower bound (Section 5.7), without attempting to deanonymise any wallet.

9.4 Data and Code Availability

The complete analysis code and per-market panel artifacts are available at https://github.com/philippdubach/polymarket-microstructure and archived at Zenodo under DOI 10.5281/zenodo.19811426. The 30-billion-event raw orderbook archive (623.8 GB) is not redistributed in the replication package due to size; access can be arranged via the corresponding author. The on-chain `OrderFilled` scrape is reproducible from the `scripts/scrape_onchain_fills.py` pipeline against any Polygon RPC provider with archive-node access.

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